Project Design Phase
Problem – Solution Fit

Date	02-10-2026
Team ID	23KD1A4233
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

Solution Fit

The ServiceNow-Powered Commuter Transit Platform replaces friction-heavy, paper-based transit operations with an agile, cloud-native ticketing engine. Designed around an intuitive Service Catalog UI, the system allows passengers to select routes, set travel dates, and configure passenger counts in a seamless, self-service web environment.

Under the hood, dynamic client-side scripting and server-side logic instantly evaluate route distances against fare matrices, delivering error-free ticket pricing in real time. Submitting an order immediately provisions a structured Requested Item (RITM) record and triggers automated event handlers that generate encrypted, single-use QR code passes attached directly to the user's booking record and sent via email notifications.

For transit operations teams, the platform consolidates system management into an executive control panel. Administrators maintain full lifecycle control over station indices, fare schedules, and route networks, while built-in analytics dashboards deliver continuous metrics on peak commuter traffic, daily booking volumes, and revenue distribution. By combining automated workflow orchestration with cloud-backed scalability, the solution slashes boarding queue delays, eliminates manual fare errors, and establishes an enterprise-grade framework for modern public transit infrastructure.

Solution Features

1. CUSTOMER SCENARIO / BENEFITS — CS

- Book metro tickets online through Service Catalog [cite: 1]
- Select source & destination stations, journey date and passenger count [cite: 1]
- User-friendly and simple interface [cite: 1]
- Automated fare calculation [cite: 2]
- Reduces manual work and errors [cite: 2]

2. KEY SERVICENOW COMPONENTS — CC

- Service Catalog (Metro Ticket Booking) [cite: 1]
- Request record creation [cite: 2]
- Flow Designer for automated workflow [cite: 2]
- Notification to user (booking confirmation) [cite: 2]
- QR code generation for digital ticket [cite: 3]
- Integration with custom tables (stations, fares) [cite: 4]

3. FEATURES & FUNCTIONALITY — AS

- Source, destination, journey date, passenger count [cite: 1]
- Fare calculation based on route information [cite: 2]
- Digital ticket generation [cite: 3]
- QR code for contactless verification [cite: 3]
- Create request record in ServiceNow [cite: 2]
- Send booking confirmation email/notification [cite: 2]

User Experience

4. USER JOURNEY — JEP

- Open Service Catalog > Metro Ticket Booking [cite: 1]
- Enter source, destination, date & passengers [cite: 1]
- View fare details (auto-calculated) [cite: 2]
- Submit request and get confirmation [cite: 2]
- Receive digital ticket with QR code [cite: 3]
- Show QR code at metro gate for entry [cite: 3]

5. PROBLEM SOLVED — RC

- Eliminates long queues at ticket counters [cite: 1]
- Removes manual fare calculation errors [cite: 2]
- Reduces waiting time for passengers [cite: 3]
- Provides a seamless and convenient experience [cite: 1]
- Improves operational efficiency [cite: 4]
- Scalable for future metro expansion [cite: 4]

6. ALTERNATIVES / LIMITATIONS — BE

- Manual ticketing (traditional counters) [cite: 1]
- Third-party ticketing apps [cite: 5]
- Limited to metro transport only (not buses/trains) [cite: 4]
- Requires internet connectivity [cite: 1]
- Depends on accurate station and fare data [cite: 4]
- QR code scanning device needed at gates [cite: 3]

Admin & Operations

7. ADMINISTRATOR CAPABILITIES — TR

- Manage metro stations and routes [cite: 4]
- Update fare details and pricing [cite: 4]
- View and manage ticket bookings [cite: 4]
- Monitor booking trends through dashboards [cite: 4]
- Generate reports for operations and revenue [cite: 4]

8. SERVICE MANAGEMENT — SL

- Track and manage requests in ServiceNow [cite: 2]
- Automated workflow for ticket creation [cite: 2]
- Send notifications and confirmations [cite: 2]
- Handle exceptions / cancellations [cite: 4]
- Maintain service catalog and request fulfillment [cite: 1]

9. BUSINESS OUTCOMES — CH

- Improved user satisfaction [cite: 1]
- Faster and hassle-free ticket booking [cite: 3]
- Higher operational efficiency [cite: 4]
- Reduced manual effort and cost [cite: 2]
- Better visibility through reports and analytics [cite: 4]
- Supports smart and scalable metro services [cite: 4]